

Supplementary Information

Cumulative demand responses to price and income changes in Korea's emissions-intensive material sectors

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Abstract

This Supplementary Information (SI) documents the response derivations, data construction, instrument definitions, complete estimates, and robustness analyses supporting the main manuscript. Sections, tables, figures, and equations use an SI prefix and are cited at the relevant points in the main text.

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SI1. Derivation of cumulative responses

Starting from the specification in annual changes in Section 3.1 of the main manuscript, the responses at each horizon are derived as follows. The estimating equation is reproduced here to define the notation.

$$\Delta \ln q_t = \alpha + \phi \Delta \ln q_{t-1} + \beta_0 \Delta \ln P_t + \beta_1 \Delta \ln P_{t-1} + \gamma_0 \Delta \ln y_t + \gamma_1 \Delta \ln y_{t-1} + \mathbf{w}_t' \delta + u_t \quad (\text{SI1})$$

Here, q_t denotes per-capita material demand, P_t the real domestic material price, and y_t real per-capita income. The vector $\mathbf{w}_t$ contains controls for external economic activity. Only the contemporaneous price change is instrumented, using the upstream input-cost variables defined in Section SI3.

Estimating annual changes does not limit the analysis to the initial-year response. Consider a price increase that occurs in year t and remains in place thereafter. In the differenced model, this appears as a one-time change in $\Delta \ln P_t$. Its effect on demand growth is β_0 in the initial year. In the following year, the response combines the persistence of the initial effect, $\phi\beta_0$, with the lagged price effect, β_1 . The lagged demand term then carries the remaining adjustment into subsequent years, generating the following response sequence:

$$r_0^P = \beta_0, \quad r_1^P = \phi r_0^P + \beta_1, \quad r_j^P = \phi r_{j-1}^P \quad (j \geq 2).$$

Summing these annual changes through horizon h gives the cumulative change in the level of demand:

$$C_h^P = \sum_{j=0}^{h-1} r_j^P, \quad C_\infty^P = \frac{\beta_0 + \beta_1}{1 - \phi}, \quad |\phi| < 1 \quad (\text{SI2})$$

The income-response path is derived in the same way. A change in the log income level produces an initial demand-growth response of γ_0 . In the following year, the response combines persistence from the initial effect, $\phi\gamma_0$, with the lagged income effect, γ_1 . Subsequent adjustment is transmitted through the same demand-persistence parameter, producing the following sequence:

$$r_0^y = \gamma_0, \quad r_1^y = \phi r_0^y + \gamma_1, \quad r_j^y = \phi r_{j-1}^y \quad (j \geq 2).$$

Summing these annual changes through horizon h gives the cumulative income response:

$$C_h^y = \sum_{j=0}^{h-1} r_j^y, \quad C_\infty^y = \frac{\gamma_0 + \gamma_1}{1 - \phi}, \quad |\phi| < 1 \quad (\text{SI3})$$

The cumulative elasticities are evaluated at $h = 1, 3$, and 5 . When $|\phi| < 1$, the sequences converge to the converged responses shown in Equations (SI2) and (SI3).

Coefficient uncertainty is propagated through Equations (SI2) and (SI3) using the procedure described in Section SI4.

SI2. Data construction and descriptive evidence

Table SI1 defines every variable, transformation, and source. Steel and cement quantities are measured in physical units, while chemicals use shipment quantity indexes because their heterogeneous products cannot be aggregated in a common physical unit.

Table SI1. Variables, construction, and sources.

Symbol	Variable	Construction	Source
q	Quantity per capita	Apparent consumption for steel and cement or domestic chemical shipment quantity index, divided by population	World Steel Association via gcamdata, Korea Cement Association, Statistics Korea
$\tilde{q}$	Broader quantity per capita	Crude steel production or total chemical shipment quantity index, divided by population	World Steel Association via gcamdata, Statistics Korea
N	Population	Midyear population in the baseline, with a separately reported KOSIS sensitivity check	UN World Population Prospects 2022, KOSIS for the sensitivity check
y	Real income per capita	Nominal GDP divided by CPI and population	Bank of Korea, UN World Population Prospects
P	Real domestic material price	Sectoral producer price index divided by CPI	Bank of Korea
z	Input cost instruments	Foreign currency prices converted to KRW using the exchange rate and divided by CPI, domestic energy price indexes divided by CPI without exchange rate conversion, then annual differences in logarithms	Sources in Table SI2
w	External activity controls	OECD and Chinese industrial production growth series used in the model	OECD data distributed through FRED

Notes. Domestic chemical shipments exclude imports and exports. Total shipments include exports but are not necessarily equal to production because inventories can change. Apparent steel consumption is measured in crude steel equivalent units and is not a subset of domestic crude steel production. The actual baseline samples are given in Section SI2 and the source ledger.

The steel series was reused from the gcamdata file WSA_steel_prod_cons_1970_2018.csv (Bond-Lamberty et al., 2019). Despite the legacy filename, the file contains annual columns for 1970–2021 and attributes the underlying data to World Steel Association Statistical Year Books for 1980, 1989, 1999, 2000, and 2019. The chemical domestic- and export-shipment indexes are from KOSIS table DT_1F02016 (items T10 and T11, industry C20). The Bank of Korea series are GDP 200Y113/10106, CPI 901Y009/0, and sectoral producer prices 404Y014/3071AA, 404Y014/30622AA, and 404Y014/305AA. Population is from World Population Prospects 2022. Table SI2 identifies the remaining instrument and control series.

After the required transformations and lags, the baseline estimation samples contain 20 annual observations for each steel outcome over 2000 to 2019, 19 for domestic chemical shipments over 2000 to 2018, 18 for total chemical shipments over 2001 to 2018, and 17 for cement over 2005 to 2021. These effective

regression samples differ from the availability windows of the source series.

Fig. SI1 presents descriptive annual price and quantity relationships and indexed price and income series over their available display periods. These patterns are descriptive. The demand responses are estimated using the strategy defined in Section SI3.

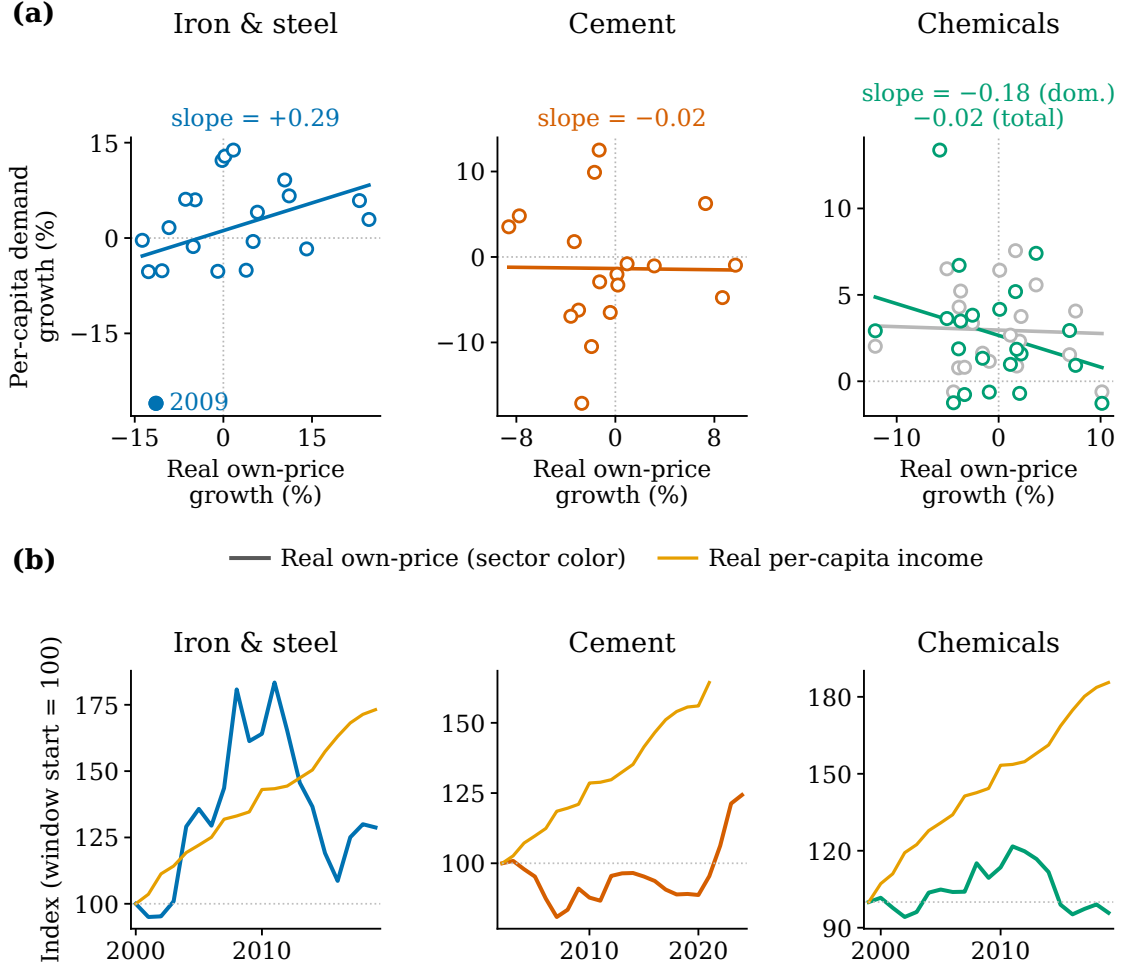


Fig. SI1. Descriptive price, quantity, and income patterns. (a) Annual log changes in quantity per capita plotted against annual log changes in the real material price, both multiplied by 100, with ordinary least squares fitted lines. The chemical panel shows domestic shipments in green and total shipments in gray. The 2009 iron and steel observation is highlighted. (b) Real material prices in the corresponding sector colors and real income per capita in orange, with each series indexed to 100 at its first displayed observation. Displayed periods follow data availability and may differ across series and from the effective estimation samples.

SI3. Instrument construction and first-stage specification

For the demand model in Equation (SI1), let z_t denote the vector of excluded upstream input-cost changes and x_t the other included regressors. The first-stage equation is:

$$\Delta \ln P_t = \pi_0 + z_t' \pi + x_t' \kappa + v_t \quad (\text{SI4})$$

The vector π measures the joint relationship between the upstream input-cost instruments and the domestic material price, conditional on the other regressors. The fitted value isolates the component of the contemporaneous price change associated with these cost movements. This component provides the price variation used to estimate the demand response in Equation (SI1).

Table SI2 provides the sectoral instrument sets, price origins, production cost channels, transformations, and sources. The selection criteria and sectoral rationale are discussed in Section 4.2 of the main manuscript.

Table SI2. Sectoral input cost instruments and proposed transmission channels.

Sector	Instrument	Market	Cost channel
Iron and steel	Australian coal benchmark price	International	Coal cost proxy
	Iron ore price	International	Blast furnace input cost
	Ferrous scrap import price	Korean imports	Electric arc furnace input cost
	Petroleum coke import unit value	Korean imports	Process fuel
	Industrial electricity price index	Domestic	Electric arc furnace power cost
Chemicals	Naphtha price, CF Japan	International	Cracker feedstock cost
	Ethylene import unit value	Korean imports	Derivative feedstock cost
	Import unit value of saturated acyclic hydrocarbons, HS 290110	Korean imports	Chemical input cost proxy
	Dubai crude oil price	International	Feedstock and fuel cost
Cement	Dubai crude oil price	International	Fuel and transport cost
	Australian coal benchmark price	International	Kiln fuel cost proxy
	Industrial electricity price index	Domestic	Grinding power cost
	Industrial city gas price index	Domestic	Plant energy cost

Notes. The coal variable used for steel and cement is the Australian benchmark PCOALAUUSDM distributed by FRED, not the Korean coal import series. Korean import unit values use UN Comtrade codes HS 7204 for ferrous scrap, HS 271311 for petroleum coke, HS 290121 for ethylene, and HS 290110 for saturated acyclic hydrocarbons. The last category is broader than ethane and can reflect changes in product composition. Iron ore and Dubai crude prices are FRED series PIORECRUSDM and POILDUBUSDM. Domestic industrial electricity and city gas prices are Bank of Korea producer price series distributed through KOSIS table DT_404Y016. Naphtha uses the CF Japan series in the Korea Chemical Industry Association raw material price table distributed through KOSIS. External activity controls are FRED series OECD-PRINTO01GYSAM and CHNPRINTO01IXPYM. Foreign currency prices are converted to KRW and divided by CPI. Domestic price indexes are divided by CPI without exchange rate conversion. Instruments enter as annual differences in logarithms.

Only the contemporaneous change in the domestic material price is instrumented; lagged demand and price changes, income growth, and the external-activity controls enter Equation (SI1) directly.

SI4. Estimation and inference procedures

Central estimates are obtained using iterated generalized method of moments (GMM), which accommodates heteroskedastic annual disturbances (Hansen, 1982). Limited information maximum likelihood (LIML) and Fuller(1) estimates assess sensitivity to the estimator in these short samples (Anderson and Rubin, 1949, Fuller, 1977). Fuller(1) applies the Fuller modification with a correction parameter of one. A coefficient ratio is reported as a converged response only when the fitted dynamics are stable.

Uncertainty in the responses after one, three, and five years and at convergence is calculated using the delta method, which carries the estimated coefficient covariance matrix through Equations (SI2) and (SI3). The main results use heteroskedasticity robust coefficient covariance estimates. For a persistent 10% increase, the exact quantity change is 100 times (1.1 raised to the cumulative elasticity minus one). Tables SI10 and SI11 apply this transformation and its derivative to the elasticity and its standard error.

Instrument strength is summarized using the conventional partial F statistic, a heteroskedasticity robust Wald statistic divided by the number of excluded instruments, and the Montiel Olea and Pflueger effective F statistic (Montiel Olea and Pflueger, 2013). The robust first stage diagnostics use HC1 covariance estimates. These measures are labelled separately and interpreted together rather than through a single mechanical threshold.

We report 90% Anderson Rubin sets for the contemporaneous price coefficient, β_0 (Anderson and Rubin, 1949, Andrews et al., 2019). Classical F inversion assumes homoskedastic disturbances. A separate HC1 Wald inversion assesses sensitivity to that assumption. These procedures concern the initial price response rather than the nonlinear cumulative responses, and need not produce the same sets. Acceptance at the edge of a finite search grid is reported explicitly and does not establish mathematical unboundedness.

Hansen's J test is reported as a joint specification diagnostic. Failure to reject is not treated as verification of the exclusion restrictions. For cement, we additionally apply the plausibly exogenous approach of Conley et al. (2012), allowing bounded direct relationships between the energy cost instruments and cement demand and reporting the corresponding range of price responses. Section SI7 provides the construction of these bounds.

Stability is assessed by estimating each specification after omitting one year or one baseline instrument at a time. Instrument omissions hold the baseline regression sample fixed. Additional checks exclude crisis observations and affected lag rows, use rolling windows of 15 years, add a second lag, and distinguish a cement population source change from the extension through 2022. The tables retain sample sizes, price and income responses, and dynamic stability flags. These exercises assess historical sensitivity, not the stability of future demand responses.

SI5. Full estimates and identification diagnostics

Tables SI3 to SI6 and Fig. SI2 report the model estimates, estimator comparisons, and identification and stability diagnostics.

Table SI3. First stage estimates for the excluded input cost instruments.**Apparent steel consumption**

Instrument	Coefficient	HC1 SE	Normal p
Australian coal benchmark price	+0.030	0.082	0.714
Iron ore price	+0.105	0.072	0.141
Ferrous scrap import price	+0.342	0.126	0.007
Industrial electricity price	−1.024	0.416	0.014
Petroleum coke import price	−0.010	0.047	0.825

Crude steel production

Instrument	Coefficient	HC1 SE	Normal p
Australian coal benchmark price	+0.020	0.076	0.797
Iron ore price	+0.093	0.085	0.272
Ferrous scrap import price	+0.399	0.149	0.008
Industrial electricity price	−0.809	0.344	0.019
Petroleum coke import price	−0.016	0.034	0.638

Chemical domestic shipments

Instrument	Coefficient	HC1 SE	Normal p
Naphtha price, CF Japan	+0.279	0.144	0.053
Ethylene import price	+0.035	0.041	0.393
Saturated acyclic hydrocarbon import unit value	+0.016	0.017	0.347
Dubai crude oil price	−0.036	0.095	0.707

Chemical total shipments

Instrument	Coefficient	HC1 SE	Normal p
Naphtha price, CF Japan	+0.231	0.134	0.084
Ethylene import price	+0.067	0.034	0.050
Saturated acyclic hydrocarbon import unit value	+0.006	0.013	0.668
Dubai crude oil price	−0.021	0.096	0.829

Cement apparent consumption

Instrument	Coefficient	HC1 SE	Normal p
Dubai crude oil price	+0.257	0.170	0.130
Australian coal benchmark price	+0.034	0.075	0.652
Industrial electricity price	+1.488	0.866	0.086
Industrial city gas price	−0.279	0.235	0.234

Outcome	N	Residual df	Conventional partial F	Partial R ²	HC1 Wald per instrument	HC1 effective F
Apparent steel consumption	20	8	8.17	0.836	21.72	9.75
Crude steel production	20	8	8.63	0.844	13.09	8.98
Chemical domestic shipments	19	8	9.27	0.823	20.58	18.13
Chemical total shipments	18	7	9.21	0.840	24.97	19.74
Cement apparent consumption	17	6	1.60	0.517	1.30	1.41

Notes. The dependent variable is the annual change in the logarithm of the real domestic material price. All equations include the controls and lagged regressors in the demand equation. Standard errors use HC1 covariance with the finite sample factor N divided by N minus the first stage parameter count. The p values use a standard normal reference distribution. The conventional partial F uses a homoskedastic covariance assumption. HC1 Wald per instrument and the Montiel Olea and Pflueger effective F are distinct statistics. Residual df refers to the unrestricted first stage. The coal instrument is an Australian benchmark and the hydrocarbon instrument is the HS290110 import unit value.

Table SI4. Full GMM estimates for all five demand specifications.

	Regressor	Apparent steel	Crude steel	Chemical domestic	Chemical total	Cement apparent
		consumption	production	shipments	shipments	consumption
8-15	Current price change	−0.011 (0.080)	+0.147 (0.046)	−0.127 (0.093)	−0.172 (0.044)	+0.013 (0.355)
	Lagged price change	−0.263 (0.052)	−0.133 (0.062)	−0.071 (0.129)	−0.038 (0.052)	−0.429 (0.261)
	Current income change	−0.005 (0.225)	−0.202 (0.353)	+0.350 (0.283)	+0.715 (0.300)	+0.772 (0.774)
	Lagged income change	+0.807 (0.208)	+0.371 (0.195)	−0.253 (0.262)	−0.083 (0.231)	−0.247 (0.551)
	Lagged quantity change	−0.004 (0.030)	+0.355 (0.092)	+0.279 (0.166)	−0.421 (0.234)	+0.113 (0.173)
	OECD industrial production	+0.015 (0.001)	+0.006 (0.001)	−0.002 (0.001)	−0.001 (0.001)	+0.002 (0.002)
	China industrial production	+0.005 (0.002)	+0.001 (0.001)	+0.000 (0.001)	+0.002 (0.001)	+0.000 (0.002)
	Constant	−0.552 (0.168)	−0.111 (0.115)	+0.045 (0.131)	−0.173 (0.072)	−0.056 (0.256)
	N	20	20	19	18	17
	Hansen J p	0.574	0.757	0.546	0.283	0.425

Notes. Heteroskedasticity robust GMM standard errors appear in parentheses. Only the contemporaneous price change is instrumented. The other regressors enter directly under the maintained exogeneity and predeterminedness assumptions. The crude steel production equation is a market boundary comparison. Failure to reject Hansen J does not establish instrument validity in these short samples.

Table SI5. Cumulative price responses and estimator stability.

	Outcome	Estimator	Effective sample	N	Converged price response and SE	Persistence	Dynamics	GMM iterations
6-15	Apparent steel consumption	GMM	2000 to 2019	20	−0.272 (0.107)	−0.004	Convergent	8
	Apparent steel consumption	LIML	2000 to 2019	20	−0.174 (0.133)	−0.078	Convergent	Not applicable
	Apparent steel consumption	Fuller(1)	2000 to 2019	20	−0.170 (0.130)	−0.077	Convergent	Not applicable
	Crude steel production	GMM	2000 to 2019	20	+0.022 (0.146)	+0.355	Convergent	7
	Crude steel production	LIML	2000 to 2019	20	−0.036 (0.162)	+0.073	Convergent	Not applicable
	Crude steel production	Fuller(1)	2000 to 2019	20	−0.034 (0.161)	+0.073	Convergent	Not applicable
	Chemical domestic shipments	GMM	2000 to 2018	19	−0.276 (0.260)	+0.279	Convergent	23
	Chemical domestic shipments	LIML	2000 to 2018	19	−0.298 (0.337)	+0.318	Convergent	Not applicable
	Chemical domestic shipments	Fuller(1)	2000 to 2018	19	−0.265 (0.318)	+0.309	Convergent	Not applicable
	Chemical total shipments	GMM	2001 to 2018	18	−0.148 (0.061)	−0.421	Convergent	35
	Chemical total shipments	LIML	2001 to 2018	18	−0.065 (0.137)	−0.077	Convergent	Not applicable

Table SI5

Outcome	Estimator	Effective sample	N	Converged price response and SE	Persistence	Dynamics	GMM iterations
Chemical total shipments	Fuller(1)	2001 to 2018	18	−0.048 (0.129)	−0.095	Convergent	Not applicable
Cement apparent consumption	GMM	2005 to 2021	17	−0.469 (0.496)	+0.113	Convergent	8
Cement apparent consumption	LIML	2005 to 2021	17	No converged response	−4.377	Nonconvergent	Not applicable
Cement apparent consumption	Fuller(1)	2005 to 2021	17	−0.095 (2.854)	−0.045	Convergent	Not applicable

Notes. Entries are cumulative log response coefficients with robust delta method standard errors in parentheses. A converged response requires the absolute value of persistence to be less than one. LIML denotes limited information maximum likelihood. Fuller(1) uses a correction parameter of one. Dynamic convergence is distinct from numerical convergence of the estimator. Baseline GMM fits used an iteration limit of 100 and did not reach it.

The cement LIML coefficient ratio is +4.297 (223.145), with persistence −4.377 and denominator 5.377. It is an unstable algebraic ratio rather than a converged demand response. Its denominator is not close to zero. Weak identification can make LIML coefficients unstable in a small sample.

Fig. SI2 compares converged cumulative price responses estimated by IV GMM and ordinary least squares (OLS) using the same dynamic specification and observations within each outcome. Under OLS, the contemporaneous price change enters the equation directly. Under IV GMM, it is instrumented using the sectoral input cost variables.

For the four outcomes shown in Fig. SI2, IV GMM gives more negative point estimates than OLS. The converged response changes from −0.14 to −0.27 for apparent steel consumption, −0.08 to −0.28 for domestic chemical shipments, +0.07 to −0.15 for total chemical shipments, and −0.33 to −0.47 for cement. This direction is consistent with positive demand shocks making OLS price responses less negative, but the comparison is not a test of endogeneity or instrument validity. The direction is not universal across outcome definitions. For crude steel production, the OLS and IV GMM estimates are −0.02 and +0.02, respectively.

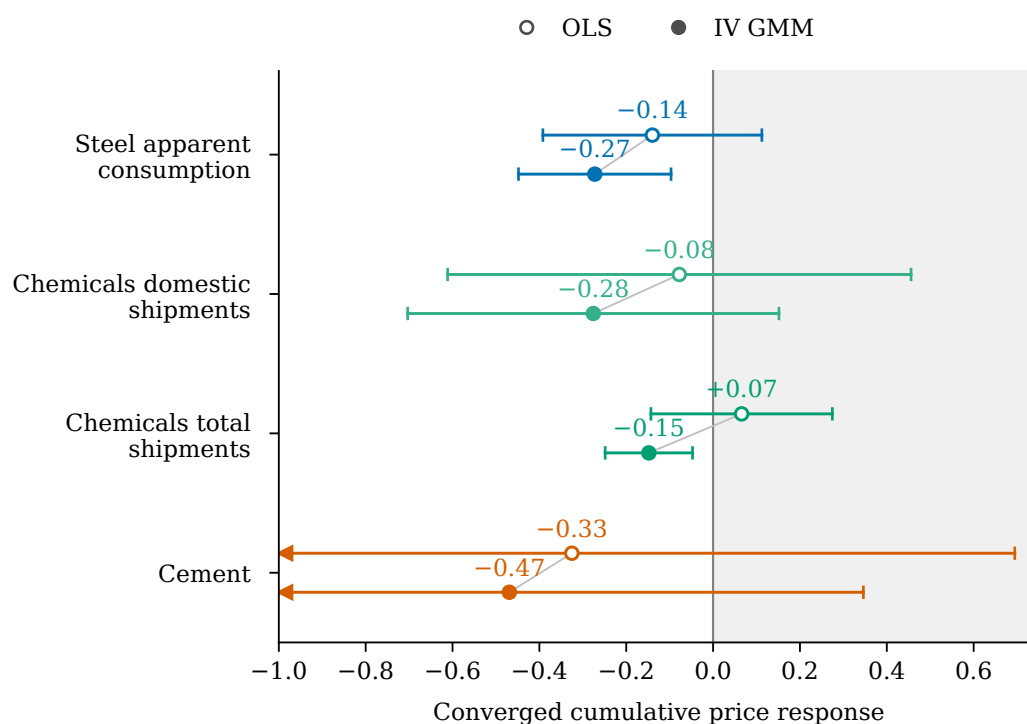


Fig. SI2. Converged cumulative price responses estimated by OLS and IV GMM. Open markers denote OLS estimates and filled markers denote IV GMM estimates. Within each outcome, both estimators use the same observations and dynamic specification. Horizontal lines show conventional 90% intervals based on delta method standard errors, using HC1 covariance for OLS and robust covariance for IV GMM. These intervals do not account for weak identification. Arrowheads indicate intervals extending beyond the plotted range. Chemical responses are shown separately for domestic and total shipments. The shaded region denotes positive price responses.

Table SI6 reports complementary measures of inferential uncertainty and specification stability. The Anderson Rubin sets characterize uncertainty in the contemporaneous price response without requiring strong instruments under the respective test assumptions. The omission ranges show how the converged response changes when a single regression row or baseline instrument is removed. Nonconvergent fits are identified separately and excluded from summaries of converged responses.

Table SI6. Initial price inference and converged response sensitivity

Outcome	Classical AR 90% set	HC1 AR 90% set	Year omission range	Convergent year fits	Instrument omission range	Convergent instrument fits
Apparent steel consumption	[−0.62, 0.50]	[−0.38, 0.42]	−0.385 to −0.218	20/20	−0.310 to −0.229	5/5
Crude steel production	[−0.55, 0.71]	[−0.06, 0.54]	−0.145 to +0.215	20/20	−0.065 to +0.058	5/5
Chemical domestic shipments	[−1.27, 0.43]	[−0.57, 0.17]	−1.638 to +0.164	19/19	−0.558 to −0.144	4/4
Chemical total shipments	[−0.63, 0.29]	[−0.05, 0.16]	−0.354 to +0.126	17/18	−0.272 to +0.100	4/4
Apparent cement consumption	Entire [−10, +10] grid accepted	[−10, −2.56] union [+1.98, +10], grid edges reached	−0.750 to +0.357	17/17	−0.932 to −0.343	4/4

Notes. Classical inversion assumes homoskedastic disturbances. HC1 Wald inversion is a separate heteroskedasticity robust sensitivity calculation. All sets concern the initial price coefficient. Finite grid edge acceptance does not establish mathematical unboundedness. Omission ranges are not confidence intervals. Instrument omissions retain the baseline regression sample. Year omissions remove a single regression row after constructing lags. The chemical total shipment fit omitting 2010 has persistence −1.007 and no converged response. It is distinct from the crisis specification purging 2009 and 2010, whose persistence is −1.069. Both fits remain in the diagnostic files.

Apparent steel consumption remains negative across all single year and baseline instrument omissions. Domestic chemical shipments remain negative across instrument omissions but not all year omissions. Crude steel production and total chemical shipments have both signs in each exercise. Cement has both signs across year omissions but remains negative across all four fixed sample instrument omissions. These historical sensitivity patterns do not establish stability during a future transition.

The classical 90% Anderson Rubin sets for the initial price coefficient include zero for all five outcomes. For cement, the full $[-10, +10]$ search grid is accepted. Within this grid, the HC1 Wald inversion instead accepts $[-10, -2.56]$ and $[+1.98, +10]$. Acceptance at the grid boundaries does not establish mathematical unboundedness.

SI6. Time-series properties and instrument-set robustness

Table SI7 evaluates the time series properties of the variables using augmented Dickey Fuller (ADF), Phillips Perron (PP), and Kwiatkowski, Phillips, Schmidt, and Shin (KPSS) tests. The level tests are mixed, whereas the results for first differences are more consistent with stationarity. The short series limit the power of these tests, so they provide diagnostic context rather than proof of the preferred specification.

Table SI7. Unit-root and stationarity tests for variables in levels and first differences.

Sector	Series	Level			First difference		
		ADF	PP	KPSS	ADF	PP	KPSS
Iron and steel	Apparent consumption	0.32	0.68	0.02	0.68	0.00	0.10
	Real per-capita income	0.28	0.04	0.03	0.05	0.00	0.14
	Real material price	0.56	0.95	0.03	0.40	0.01	0.24
Cement	Apparent consumption	0.15	0.21	0.09	0.01	0.01	0.39
	Real per-capita income	0.06	0.06	0.20	0.04	0.00	0.55
	Real material price	0.95	1.00	0.07	0.42	0.15	0.09
Chemicals	Total shipments	0.26	0.33	0.01	0.00	0.00	0.06
	Domestic shipments	0.59	0.76	0.07	0.00	0.00	0.74
	Real per-capita income	0.55	0.00	0.03	0.27	0.00	0.05
	Real material price	0.94	0.97	0.03	0.01	0.01	0.27

Notes. Entries are p values. Level specifications include a constant and linear trend; first-difference specifications include a constant. Small p values reject a unit root under the ADF and PP tests but reject stationarity under the KPSS test.

Table SI8 complements these tests with the bounds test of Pesaran et al. (2001), which examines whether demand, income, and price share a stable relationship in levels. The reported F statistics fall below the lower critical bound. These results provide no evidence for imposing a levels relationship in the selected specification, although the short samples limit the power of the test.

Table SI8. Pesaran–Shin–Smith bounds tests for a level relationship among quantity, income, and price.

Outcome	Bounds F statistic
Apparent steel consumption	0.62
Chemical domestic shipments	2.40
Chemical total shipments	1.28
Cement apparent consumption	2.87

Notes. Each specification includes an unrestricted intercept and two level regressors ($k = 2$). The $I(0)$ lower critical bounds are 3.17 at the 10% level and 3.79 at the 5% level. All reported F statistics fall below both bounds, so the null hypothesis of no level relationship is not rejected.

Table SI9 examines sensitivity to the composition of the reported baseline instrument set while holding the regression sample and included regressors fixed.

Table SI9. All overidentified subsets of the baseline instrument sets

Outcome	Subsets fitted	Usable convergent fits	Negative responses	Converged price range	N
Apparent steel consumption	26	26	26/26	−0.316 to −0.045	20
Crude steel production	26	26	22/26	−0.094 to +0.124	20
Chemical domestic shipments	11	11	11/11	−0.564 to −0.058	19
Chemical total shipments	11	11	7/11	−0.272 to +0.110	18
Apparent cement consumption	11	11	10/11	−2.824 to +0.268	17

Notes. Each outcome uses every subset containing at least two of its reported baseline instruments, with included regressors and regression years held fixed. All 85 subset fits have stable dynamics and terminate below the iteration cap. The sign shares describe this finite set of specifications, not probabilities or confidence levels. The submitted broader candidate pools are not used in this replacement design. Full coefficients, covariances, income responses, instrument identities, years and solver diagnostics are available in the associated audit files.

SI7. Additional identification checks and cumulative response paths

Cement provides substantially less first stage information than iron and steel or chemicals. In the baseline specification, the conventional partial F statistic is 1.60, the HC1 Wald statistic per excluded instrument is 1.30, and the HC1 Montiel Olea and Pflueger effective F statistic is 1.41. The analysis below examines sensitivity to bounded direct effects of the instruments.

SI7.1. Cement identification checks

Table SI12 relaxes the strict exclusion restriction by allowing each energy cost instrument to have a bounded direct effect on cement demand. Under nonpositive direct effects and $c = 0.05$, the conditional price response range is -0.548 to -0.367 . Its endpoints are unchanged to three decimals across grids with five, seven, and nine points per instrument. The corresponding conventional 90% envelope is -1.331 to $+0.412$. At $c = 0.10$, the endpoints are sensitive to grid resolution and are reported as exploratory numerical results rather than precise bounds. These conditional ranges and conventional envelopes are not weak instrument robust confidence sets.

SI7.2. Cumulative quantity-response paths

Tables SI10 and SI11 translate the estimated elasticities into exact percentage changes in each quantity measure following a permanent 10% increase in the real domestic material price or real income per capita. The responses after one, three, and five years and at convergence are generated from the same fitted dynamics rather than estimated independently at each horizon.

Table SI10. Cumulative quantity responses to a permanent 10% increase in the real domestic material price.

Outcome	1 year	3 years	5 years	Converged
Apparent steel consumption	-0.11 (0.76)	-2.56 (0.99)	-2.56 (0.99)	-2.56 (0.99)
Crude steel production	+1.41 (0.45)	+0.36 (1.25)	+0.23 (1.38)	+0.21 (1.40)
Chemical domestic shipments	-1.21 (0.88)	-2.49 (2.22)	-2.59 (2.39)	-2.59 (2.41)
Chemical total shipments	-1.62 (0.41)	-1.44 (0.49)	-1.41 (0.55)	-1.40 (0.57)
Cement apparent consumption	+0.12 (3.39)	-4.32 (4.55)	-4.37 (4.52)	-4.37 (4.52)

Notes. Entries are percentage changes in quantity, calculated as 100 times the difference between 1.1 raised to the cumulative response coefficient and one. Delta method standard errors appear in parentheses and use the derivative of this transformation. They are not obtained by multiplying the coefficient and its standard error by ten. Responses after three and five years follow the estimated dynamics rather than separate horizon specific regressions. The standard errors are conventional GMM delta method quantities and are not robust to weak instruments.

Table SI11. Cumulative quantity responses to a permanent 10% increase in real income per capita.

Outcome	1 year	3 years	5 years	Converged
Apparent steel consumption	−0.05 (2.14)	+7.91 (2.91)	+7.91 (2.91)	+7.91 (2.91)
Crude steel production	−1.90 (3.30)	+1.96 (4.06)	+2.46 (4.17)	+2.53 (4.18)
Chemical domestic shipments	+3.39 (2.79)	+1.45 (4.44)	+1.30 (4.74)	+1.29 (4.77)
Chemical total shipments	+7.05 (3.06)	+4.81 (2.65)	+4.42 (2.58)	+4.33 (2.56)
Cement apparent consumption	+7.64 (7.94)	+5.83 (13.40)	+5.81 (13.49)	+5.81 (13.49)

Notes. Entries are percentage changes in quantity, calculated as 100 times the difference between 1.1 raised to the cumulative response coefficient and one. Delta method standard errors appear in parentheses and use the derivative of this transformation. They are not obtained by multiplying the coefficient and its standard error by ten. Responses after three and five years follow the estimated dynamics rather than separate horizon specific regressions. The standard errors are conventional GMM delta method quantities and are not robust to weak instruments.

Table SI12. Sensitivity of the cement price response to bounded direct instrument effects.

Bound c	Nonpositive point range	Nonpositive conventional 90% envelope	Either sign point range	Either sign conventional 90% envelope	Grid status
0.00	[−0.469, −0.469]	[−1.284, +0.346]	[−0.469, −0.469]	[−1.284, +0.346]	Standard IV baseline
0.02	[−0.487, −0.379]	[−1.302, +0.396]	[−0.657, −0.350]	[−1.544, +0.463]	Five point grid
0.05	[−0.548, −0.367]	[−1.331, +0.412]	[−0.962, −0.223]	[−1.912, +0.787]	Five point grid
0.10	[−0.602, +1.210]	[−1.729, +4.149]	[−1.261, +0.782]	[−2.468, +2.765]	Exploratory and grid sensitive

Notes. Each direct effect coefficient links an instrument log change to a demand log change. Under the signed restriction the coefficient is between $-c$ and zero. Under the alternative restriction it is between $-c$ and c . A bound of 0.05 allows a 10% instrument price increase to reduce demand directly by approximately 0.5% under the signed restriction. The intervals summarize extrema on a grid, with GMM weights updated at each point. The point ranges are conditional on these restrictions and are not confidence intervals. The conventional 90% envelope spans the individual delta method intervals, each calculated with a multiplier of 1.645. It is not a weak instrument robust confidence set. The zero bound reproduces the standard IV estimate.

The signed point range at $c = 0.05$ is unchanged at three decimals on grids of five, seven and nine points per instrument dimension. The $c = 0.10$ endpoints vary with grid resolution and are reported only as exploratory results. They are not a reliable headline range or an established threshold for reaching zero. The grid calculations do not record iteration counts for individual fits.

SI7.3. Additional historical sensitivity checks

Table SI13. Crisis observation exclusions.

Apparent steel consumption

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
Baseline	20	12	8	−0.272 (0.107)	+0.799 (0.283)	8.17	−0.004	0.004	Convergent
Omit 2009	19	11	7	−0.279 (0.111)	+1.195 (0.433)	7.00	−0.022	0.022	Convergent
Purge 2009 and 2010	18	10	6	−0.318 (0.088)	+1.108 (0.512)	6.09	−0.014	0.014	Convergent
Omit 2008 and 2009	18	10	6	−0.403 (0.103)	+1.101 (0.437)	4.44	−0.019	0.019	Convergent
Purge 2008 to 2010	17	9	5	−0.383 (0.083)	+1.350 (0.426)	3.38	−0.119	0.119	Convergent

Crude steel production

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
Baseline	20	12	8	+0.022 (0.146)	+0.262 (0.428)	8.63	+0.355	0.355	Convergent
Omit 2009	19	11	7	−0.023 (0.128)	−0.096 (0.572)	7.36	+0.292	0.292	Convergent
Purge 2009 and 2010	18	10	6	−0.019 (0.134)	+0.035 (0.496)	8.18	+0.331	0.331	Convergent
Omit 2008 and 2009	18	10	6	−0.104 (0.151)	−0.287 (0.680)	4.89	+0.200	0.200	Convergent
Purge 2008 to 2010	17	9	5	−0.057 (0.178)	−0.048 (0.593)	5.16	+0.316	0.316	Convergent

Table SI13. Chemical domestic shipments

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
Chemical domestic shipments									
Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
Baseline	19	11	8	−0.276 (0.260)	+0.134 (0.495)	9.27	+0.279	0.279	Convergent
Omit 2009	18	10	7	−0.419 (0.225)	−0.101 (0.491)	10.29	+0.238	0.238	Convergent
Purge 2009 and 2010	17	9	6	−0.260 (0.222)	+0.152 (0.427)	18.01	+0.191	0.191	Convergent
Omit 2008 and 2009	17	9	6	−0.426 (0.218)	−0.082 (0.495)	6.32	+0.229	0.229	Convergent
Purge 2008 to 2010	16	8	5	−0.033 (0.212)	+0.346 (0.440)	9.69	+0.261	0.261	Convergent

Chemical total shipments

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
Baseline	18	10	7	−0.148 (0.061)	+0.445 (0.258)	9.21	−0.421	0.421	Convergent
Omit 2009	17	9	6	−0.178 (0.058)	+0.153 (0.187)	11.22	−0.211	0.211	Convergent
Purge 2009 and 2010	16	8	5	No converged response	No converged response	35.92	−1.069	1.069	Nonconvergent
Omit 2008 and 2009	16	8	5	+0.002 (0.077)	+0.108 (0.171)	8.15	−0.105	0.105	Convergent
Purge 2008 to 2010	15	7	4	+0.138 (0.025)	+0.213 (0.086)	20.76	−0.745	0.745	Convergent

Table SI13. Cement apparent consumption

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
Cement apparent consumption									
Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
Baseline	17	9	6	−0.469 (0.496)	+0.592 (1.338)	1.60	+0.113	0.113	Convergent
Omit 2009	16	8	5	−0.160 (0.488)	+1.258 (1.967)	1.16	+0.159	0.159	Convergent
Purge 2009 and 2010	15	7	4	+0.601 (0.251)	+6.057 (0.407)	0.94	−0.379	0.379	Convergent
Omit 2008 and 2009	15	7	4	+0.439 (0.409)	+4.066 (1.663)	0.63	+0.034	0.034	Convergent
Purge 2008 to 2010	14	6	3	+0.285 (0.201)	+5.876 (0.468)	0.52	−0.515	0.515	Convergent
Omit 2020 and 2021	15	7	4	+1.262 (0.228)	+6.854 (0.757)	0.47	−0.499	0.499	Convergent

Notes. Price and income entries are converged cumulative log response coefficients with heteroskedasticity robust delta method standard errors. Equation and first stage df are residual degrees of freedom after counting all parameters. Dynamic convergence requires every autoregressive companion root to lie inside the unit circle. Numerical iteration counts are not recorded in the source for these supplementary fits, so the dynamics column does not certify numerical optimizer convergence. Complete years and diagnostics are retained in the accompanying analysis files.

Omission drops the specified equation rows after constructing lags. Purging also removes the next equation when it contains a lagged crisis observation. The chemical

total shipment specification purging 2009 and 2010 has persistence -1.069 and no converged price or income response. Its algebraic coefficient ratios remain in the machine readable diagnostic file but are not interpreted as demand responses. Crisis checks assess sensitivity within the observed sample and are not formal structural break tests.

Table SI14. Rolling windows spanning 15 years.

Apparent steel consumption

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
2000 to 2014	15	7	3	-0.319 (0.039)	$+0.664$ (0.148)	5.16	$+0.005$	0.005	Convergent
2001 to 2015	15	7	3	-0.292 (0.133)	$+0.654$ (0.542)	2.76	$+0.006$	0.006	Convergent
2002 to 2016	15	7	3	$+0.163$ (0.078)	-0.162 (0.314)	3.33	-0.056	0.056	Convergent
2003 to 2017	15	7	3	-0.257 (0.103)	$+0.712$ (0.342)	6.46	$+0.015$	0.015	Convergent
2004 to 2018	15	7	3	-0.422 (0.031)	$+0.412$ (0.324)	5.30	$+0.012$	0.012	Convergent
2005 to 2019	15	7	3	-0.365 (0.058)	$+0.423$ (0.341)	8.56	$+0.041$	0.041	Convergent

Crude steel production

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
2000 to 2014	15	7	3	+0.089 (0.086)	+0.118 (0.264)	6.77	−0.057	0.057	Convergent
2001 to 2015	15	7	3	−0.207 (0.157)	−0.002 (0.787)	3.98	+0.268	0.268	Convergent
2002 to 2016	15	7	3	−0.262 (0.197)	−1.256 (1.050)	3.11	+0.238	0.238	Convergent
2003 to 2017	15	7	3	−0.087 (0.129)	−1.583 (0.881)	7.90	+0.157	0.157	Convergent
2004 to 2018	15	7	3	−0.118 (0.083)	−2.340 (0.526)	4.49	−0.100	0.100	Convergent
2005 to 2019	15	7	3	+0.442 (0.080)	−0.420 (0.386)	3.91	+0.190	0.190	Convergent

Table SI14. Chemical domestic shipments

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
Chemical domestic shipments									
Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
2000 to 2014	15	7	4	−1.454 (1.534)	−3.220 (4.014)	3.58	+0.809	0.809	Convergent
2001 to 2015	15	7	4	−0.656 (0.736)	+0.083 (1.504)	4.42	+0.741	0.741	Convergent
2002 to 2016	15	7	4	−1.380 (1.128)	−1.231 (1.476)	11.12	+0.703	0.703	Convergent
2003 to 2017	15	7	4	−0.114 (1.808)	+8.921 (20.792)	16.11	+0.906	0.906	Convergent
2004 to 2018	15	7	4	−3.525 (7.356)	+8.988 (23.559)	11.95	+0.879	0.879	Convergent

Chemical total shipments

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
2001 to 2015	15	7	4	−0.266 (0.085)	−0.305 (0.215)	3.87	+0.152	0.152	Convergent
2002 to 2016	15	7	4	−0.364 (0.144)	−0.290 (0.244)	13.33	+0.010	0.010	Convergent
2003 to 2017	15	7	4	+0.140 (0.098)	+0.983 (0.406)	14.43	+0.455	0.455	Convergent
2004 to 2018	15	7	4	−0.341 (0.122)	−0.095 (0.434)	12.59	+0.382	0.382	Convergent

Cement apparent consumption

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
2005 to 2019	15	7	4	+1.262 (0.228)	+6.854 (0.757)	0.47	−0.499	0.499	Convergent
2006 to 2020	15	7	4	−0.218 (0.319)	+0.111 (1.291)	9.35	+0.136	0.136	Convergent
2007 to 2021	15	7	4	+0.410 (0.551)	+2.192 (1.622)	24.17	+0.404	0.404	Convergent

Notes. Price and income entries are converged cumulative log response coefficients with heteroskedasticity robust delta method standard errors. Equation and first stage df are residual degrees of freedom after counting all parameters. Dynamic convergence requires every autoregressive companion root to lie inside the unit circle. Numerical iteration counts are not recorded in the source for these supplementary fits, so the dynamics column does not certify numerical optimizer convergence. Complete

years and diagnostics are retained in the accompanying analysis files.

Adjacent windows overlap and are not independent replications. Small first stage degrees of freedom limit the precision of this exercise. For cement, the 2005 to 2019 window contains the same observations as the specification omitting 2020 and 2021.

Table SI15. Lag order sensitivity.

Apparent steel consumption

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
One lag	20	12	8	−0.272 (0.107)	+0.799 (0.283)	8.17	−0.004	0.004	Convergent
Two lags	20	9	5	−0.314 (0.085)	+2.318 (0.380)	16.87	−0.419	0.504	Convergent

Crude steel production

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
One lag	20	12	8	+0.022 (0.146)	+0.262 (0.428)	8.63	+0.355	0.355	Convergent
Two lags	20	9	5	−0.196 (0.104)	−0.058 (0.368)	9.21	+0.101	0.534	Convergent

Chemical domestic shipments

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
One lag	19	11	8	−0.276 (0.260)	+0.134 (0.495)	9.27	+0.279	0.279	Convergent
Two lags	19	8	5	−0.441 (0.271)	−0.518 (0.531)	5.98	+0.238	0.439	Convergent

Chemical total shipments

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
One lag	18	10	7	−0.148 (0.061)	+0.445 (0.258)	9.21	−0.421	0.421	Convergent
Two lags	17	6	3	−7.541 (24.764)	−10.316 (35.543)	7.28	+0.891	0.913	Convergent

Cement apparent consumption

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
One lag	17	9	6	−0.469 (0.496)	+0.592 (1.338)	1.60	+0.113	0.113	Convergent
Two lags	16	5	2	+0.367 (0.212)	+4.549 (0.796)	30.90	−0.402	0.590	Convergent

Notes. Price and income entries are converged cumulative log response coefficients with heteroskedasticity robust delta method standard errors. Equation and first stage df are residual degrees of freedom after counting all parameters. Dynamic convergence requires every autoregressive companion root to lie inside the unit circle. Numerical iteration counts are not recorded in the source for these supplementary fits, so the dynamics column does not certify numerical optimizer convergence. Complete years and diagnostics are retained in the accompanying analysis files.

The two lag specification adds one lag each of quantity, price and income while retaining the baseline controls and excluded instruments. Effective sample sizes can change when additional lags require unavailable observations. Chemical total shipments remain dynamically convergent, but the two lag price response is imprecise. Its persistence sum of 0.891 leaves a long run denominator of 0.109. This case differs from the nonconvergent crisis specification.

Table SI16. Cement population source and sample extension.

Cement apparent consumption

Specification	N	Equation df	First stage df	Price and SE	Income and SE	Conventional partial F	Persistence sum	Largest AR root modulus	Dynamics
UN population, 2005 to 2021	17	9	6	−0.469 (0.496)	+0.592 (1.338)	1.60	+0.113	0.113	Convergent
KOSIS population, 2005 to 2021	17	9	6	−0.238 (0.199)	+4.831 (0.592)	2.14	−0.437	0.437	Convergent
KOSIS population, 2005 to 2022	18	10	7	+0.218 (0.195)	+4.654 (0.636)	3.08	−0.277	0.277	Convergent

Notes. Price and income entries are converged cumulative log response coefficients with heteroskedasticity robust delta method standard errors. Equation and first stage df are residual degrees of freedom after counting all parameters. Dynamic convergence requires every autoregressive companion root to lie inside the unit circle. Numerical iteration counts are not recorded in the source for these supplementary fits, so the dynamics column does not certify numerical optimizer convergence. Complete years and diagnostics are retained in the accompanying analysis files.

The population change is applied consistently to quantity and income per capita. The matched sample comparison separates the population source change from adding 2022. Other required variables limit the available extension to 2022. Each reported conventional 90% price interval includes zero.

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Appendix. Published benchmarks for material demand responses

Table A1. Published benchmarks for material demand responses.

Study	Outcome	Income or		Interpretation
		Price response	activity response	
This study	Korean apparent steel consumption	−0.272	+0.799	Converged GMM response
Fernandez (2018a)	World steel consumption	−0.086	+1.156	IV system in log levels
Tcha and Takashina (2002)	World steel consumption	−0.012	+1.030	Contemporaneous differences with a constant and Divisia aggregates
Crompton (2015)	Steel consumption in 26 OECD countries	Not estimated	0.01 to 4.05	Country income elasticities from a static panel
This study	Korean apparent cement consumption	−0.548 to −0.367	Not part of this set	Conditional point response set under the signed bound of 0.05
Seo (2021)	Korean domestic cement shipments	0.842 total cement and 0.977 Portland cement	Not estimated	Monthly IV-ECM; absolute elasticities
Stuermer (2017)	Five base metals in 15 countries	−0.691 to −0.035	+0.428 to +1.495	Long-run PMG; manufacturing output

Notes. Estimates differ in sample, outcome, frequency, and method and do not form a pooled range. Tcha and Takashina use Divisia relative metal prices and an aggregate metal quantity index rather than GDP. Crompton reports variation across countries. Stuermer reports ranges across aluminum, copper, lead, tin, and zinc, with activity measured by real manufacturing value added per capita. Seo reports absolute magnitudes, not positive demand slopes. The present cement set assumes the signed direct-effect bound $c = 0.05$ and is not a confidence interval. Uncertainty for this study is reported in Section 5.

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